Staphyloenterotoxemia (Staphylococcal Food Poisoning): A Comprehensive Disease Characterization

Disease: Staphyloenterotoxemia (staphylococcal food poisoning; staphylococcal toxemia due to food) **MONDO ID:** MONDO:0005971 · **MeSH:** Staphylococcal Food Poisoning (D013205) · **ICD-10:** A05.0 · **ICD-11:** bacterial foodborne intoxication · **OMIM:** not applicable (not a human Mendelian disease) · **Orphanet:** not a listed rare disease (common foodborne illness) **Category:** Infectious Disease (toxin-mediated food intoxication) **Causal "gene(s)":** *Bacterial* staphylococcal enterotoxin (*se*) genes — **sea, seb, sec, sed, see** and >20 newer *se/sel* types — carried by *Staphylococcus aureus* (NCBI Taxon:1280). **There is no human causal gene** and no OMIM Mendelian entry.

Summary

Staphyloenterotoxemia is not a genetic disease and not an active bacterial infection — it is an acute intoxication caused by ingesting preformed, heat-stable staphylococcal enterotoxins (SEs) that *Staphylococcus aureus* has already produced in mishandled food. The "causal genes" named in the research template refer to *bacterial* enterotoxin genes (*sea*, *seb*, *sec*, *sed*, *see*, and a large family of newer *se/sel* genes carried on prophages, pathogenicity islands, plasmids, and the *egc* cluster), not any human gene. There is no OMIM entry, no inheritance pattern, no human susceptibility locus, and no genetic test of the patient; the disease is diagnosed epidemiologically and by direct detection of toxin or *se* genes in food/isolates. SEA is the enterotoxin most frequently implicated in human outbreaks.

The clinical syndrome is a short-incubation emetic gastroenteritis: nausea, forceful repeated vomiting, retching, and abdominal cramps begin roughly 1–6 hours after ingestion (typically 2–4 h), often with variable non-bloody diarrhea and usually without high fever, and resolve spontaneously within 24–48 hours. The short incubation is the signature of a *preformed* toxin acting directly, in contrast to the ~12–24 h incubation of most live-pathogen foodborne infections. Because the toxin is heat-stable, ordinary cooking/reheating does not inactivate it,

and the toxin persists in food even after the bacteria are gone — so control must prevent toxin *formation* through hygiene and temperature management, and treatment of the ill patient is purely supportive (rehydration ± antiemetics; antibiotics are not indicated).

Mechanistically the disease has **two separable arms**. The dominant, clinically defining arm is an **emetic reflex**: enterotoxin binds intestinal submucosal mast cells, triggering degranulation and release of serotonin (5-HT) and histamine, which activate 5-HT3 receptors on vagal afferent neurons to drive the vomiting reflex — a pathway proven by vagotomy and 5-HT3-antagonist experiments in animal models. The second arm is the **superantigen** property of SEs: they cross-link MHC class II on antigen-presenting cells with T-cell-receptor Vβ domains, causing polyclonal T-cell activation and a pro-inflammatory cytokine storm (IL-2, IFN-γ, TNF) that underlies toxic-shock-like illness at high or parenteral doses. These two arms explain why the same toxin family produces both a self-limited emetic illness (oral food dose) and severe systemic superantigen pathology (high-dose/parenteral contexts).

Key Findings

F001 — Etiology: a preformed-toxin intoxication, not infection or inherited disease

Staphyloenterotoxemia is caused by ingestion of **preformed** *S. aureus* enterotoxins in food, not by colonization or an inherited defect. In the 2024 Shenzhen corporate-canteen outbreak, both *S. aureus* and enterotoxin A were recovered from patient specimens and from the implicated hand-shredded chicken, with an attack rate of 15% and a strong association with the suspect food (RR ≈ 5.91, 95% CI 1.41–24.76) (PMID: 42700799). The investigators explicitly noted that *"brief steaming before service would not have removed preformed toxin,"* underscoring that the agent is a heat-stable toxin rather than a live organism. Surveillance literature frames the general etiology plainly: *"Staphylococcal foodborne illness is a prevalent global foodborne disease caused by the ingestion of food contaminated with Staphylococcus aureus enterotoxins"* (PMID: 41331910). The "causal genes" are therefore *bacterial* enterotoxin genes; **no human Mendelian cause exists and OMIM has no entry.**

F002 — Emetic mechanism: mast-cell 5-HT/histamine → 5-HT3 vagal afferents → vomiting

The vomiting that defines the illness is a neuro-immune reflex. In the house musk shrew (*Suncus murinus*) emesis model, SEA-induced emesis was blocked by a 5-HT synthesis inhibitor and a 5-HT3 receptor antagonist, SEA increased intestinal 5-HT release, and *"SEA-induced emesis was also abolished by surgical vagotomy,"* establishing that *"the 5-HT3 receptors on vagal afferent neurons are essential for SEA-stimulated emesis"* (PMID: 17517065). In the common marmoset primate model, *"SEA bound with submucosal mast cells and induced mast cell degranulation"* and histamine release, and the classical emetic SEs (SEA/SEB/SEC) induced vomiting whereas the non-emetic superantigen TSST-1 did not (PMID: 31112582). The upstream trigger is thus mast-cell degranulation; the downstream effector is the vagal-afferent 5-HT3 emetic pathway.

F003 — Heat stability: temperature abuse drives toxin accumulation; preformed toxin survives cooking

SEs are explicitly *"heat-stable enterotoxins encoded by virulence genes sea and seb"* (PMID: 42418034). Predictive microbiology in chicken shows that *S. aureus* grows and begins producing detectable enterotoxin once populations exceed roughly 6.5–8.0 log CFU/g (e.g., at 37 °C after 8–13 h), with some growth possible as low as 10 °C in certain products (PMID: 42314251). Critically, the toxin persists in food *"even in the absence of viable microorganisms, making direct toxin detection essential"* (PMID: 42190559), and in the Shenzhen outbreak brief steaming failed to remove preformed toxin (PMID: 42700799). This is the single most important control-and-diagnosis fact: **cooking kills bacteria but not toxin.**

F004 — Superantigen arm: MHC II × TCR-Vβ cross-linking → polyclonal T-cell cytokine storm

SEs are superantigens that *"polyclonally activate T cells by cross-linking MHC class II on antigen-presenting cells with T cell receptor (TCR) Vβ regions, generating potent IL-2/IFN-γ/TNF responses"* distinct from conventional peptide-MHC-restricted activation (PMID: 41187575). SEB-type toxins are pyrogenic superantigens capable of inducing toxic shock (PMID: 30909619). Mucosa-associated invariant T (MAIT) cells are among the most powerful early cytokine producers in response to SEB and then rapidly acquire an anergic/exhausted phenotype,

defining a mechanism of superantigen-induced immunopathology followed by immunosuppression (PMID: 28632753). This arm is largely separable from the emetic reflex and dominates at high or parenteral toxin doses.

F005 — Epidemiology & prognosis: among the most common foodborne diseases, self-limited, heavily under-reported

Staphylococcal food poisoning is *"one of the most common Foodborne Diseases (FBD) in the world,"* with *"underreporting caused by the short duration of clinical symptoms and lack of medical care"* (PMID: 36809532). It is described as a *"leading foodborne illness"* (PMID: 42220823) and a *"prevalent global foodborne disease"* (PMID: 41331910). Outbreaks are characteristically point-source with high attack rates — e.g., 16 of 107 exposed employees (attack rate 15.0%) in the Shenzhen canteen outbreak (PMID: 42700799).

F006 — Food vehicles and reservoirs: post-cooking-handled protein/dairy foods; colonized handlers and mastitic milk

Implicated vehicles include hand-shredded cooked chicken (PMID: 42700799), cheese and dairy (PMID: 42220823), retail meats and camel meat/offal (PMID: 42376519), and milk from animals with subclinical mastitis, where PCR detected enterotoxin genes in coagulase-positive staphylococci (PMID: 42376077). Food handlers are a major reservoir: *"Food handlers can contain these dangerous bacteria, which can contaminate food and cause staphylococcal food poisoning"* (PMID: 41331910), and an asymptomatic colonized food handler was identified in the Shenzhen outbreak (PMID: 42700799).

F007 — Causal "genes": the staphylococcal enterotoxin (*se*) gene family, SEA predominating in human outbreaks

The classical enterotoxins SEA–SEE (genes *sea*, *seb*, *sec*, *sed*, *see*) plus >20 newer *se/sel* types (e.g., *seg–selu*, *seh*, *ser*) constitute the "causal gene" set. In a genomic study of foodborne-outbreak *S. aureus* isolates, *"all strains produced classical staphylococcal enterotoxins, with staphylococcal enterotoxin (SEA) showing the highest detection rate (63.16%)"* (PMID: 42350958). Reservoir-specific prevalence differs: in spent laying hens, *"503 of 995 isolates (50.6%) carried one or more staphylococcal enterotoxin (SE) genes, with seh being the most prevalent, followed by sed and ser"* (PMID: 42425691); in buffalo subclinical-mastitis isolates, *sec* (8.5%), *sea* (8.2%), *sed* (4.5%), *seb* (2.8%), and *see* (1.7%) were detected (PMID: 42406231).

F008 — Diagnosis: epidemiology + direct toxin detection and *se*-gene PCR/ WGS, not host testing

Because toxin persists without viable bacteria, *"direct toxin detection [is] essential for food safety surveillance"* (PMID: 42190559). Enterotoxin is quantified by ELISA on culture supernatants and in food matrices (*"we quantified production of selected enterotoxins in culture supernatants… by enzyme-linked immunosorbent assay (ELISA)"*) (PMID: 42425691), and outbreak investigation combines culture, immunoassay toxin detection, WGS virulence-gene profiling, and MLST (PMID: 42350958; PMID: 42700799). Emerging rapid platforms (e.g., aptasensors for *S. aureus*) are in development (PMID: 42967785). **No patient genetic/host test is used.**

F009 — Animal models: emesis bioassays plus natural disease in food animals

Emetic activity is studied in toxin-challenge (non-genetic) models: the house musk shrew reproduces SEA-induced emesis via intestinal 5-HT/5-HT3/vagal signaling (PMID: 17517065), and the common marmoset is a primate emetic model in which *"Common marmosets were administered classical SEs, including SEA, SEB and SEC, and exhibited multiple vomiting responses. However, a non-emetic staphylococcal superantigen, toxic shock syndrome toxin-1, did not induce emesis"* (PMID: 31112582). Rhesus/cynomolgus monkeys and kittens are historical emetic bioassays. Enterotoxigenic *S. aureus* naturally colonizes food-producing animals — *"S. aureus was detected in 649 of 800 birds (81.1%)"* in spent laying hens (PMID: 42425691) — and causes bovine/buffalo subclinical mastitis carrying *se* genes (PMID: 42376077; PMID: 42406231).

F010 — Prevention: food-safety control of toxin formation; no vaccine or genetic prophylaxis

Primary prevention targets the two preventable failures seen in outbreaks: *"bare-hand contact with cooked, ready-to-eat food, and inadequate cooling of large batches during storage"* (PMID: 42700799). Because preformed heat-stable toxin survives reheating and persists without viable bacteria (PMID: 42190559), control must prevent toxin *formation* via temperature control (rapid cooling, cold holding), supported by predictive growth/toxin models (PMID: 42314251). Upstream biocontrol is experimental: a *Lactobacillus/Bifidobacterium* consortium achieved a *"4.56 log_{10} CFU/mL reduction and 58.1% enterotoxin suppression at 72 h"* with *sea*

downregulation in vitro (PMID: 42418034), and lytic bacteriophage UHP46 reduced *S. aureus* in dairy matrices (PMID: 42782413). **There is no human vaccine or pharmacologic prophylaxis.**

F011 — Clinical phenotype: acute, short-incubation (~1–6 h) emetic gastroenteritis, self-limited in 24–48 h

SEs are *"thermostable superantigenic proteins produced by Staphylococcus aureus that are responsible for staphylococcal food poisoning (SFP) in humans"* (PMID: 42625416). The hallmark is rapid onset 0.5–6 h (typically 2–4 h) after ingestion, dominated by nausea, forceful repeated vomiting, retching, and abdominal cramping, often with non-bloody diarrhea; fever is usually absent or low-grade. The short incubation reflects the action of *preformed* toxin (vs ~12–24 h for most live-pathogen foodborne infections). Animal data mechanistically explain the emesis-dominant phenotype (PMID: 31112582; PMID: 17517065).

F012 — Treatment: supportive only; antibiotics not indicated

Because illness results from preformed toxin rather than active infection, eradicating bacteria is irrelevant to the acute episode and antibiotics do not alter its course; management is symptomatic (fluid/electrolyte replacement ± antiemetics). The self-limited course — *"the short duration of clinical symptoms"* (PMID: 36809532) — means supportive care suffices for the typical oral-dose illness. No toxin-specific antidote or vaccine is approved; biocontrol (PMID: 42418034) and phage (PMID: 42782413) are food-stage interventions, not human therapeutics.

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

```
1. S. aureus (carrying se gene[s]) contaminates protein-/dairy-rich food,
   typically via a colonized food handler's hands or mastitic animal milk.
        → leads to
2. Temperature abuse (inadequate cooling / warm holding) permits S. aureus
   growth to ~>6.5–8 log CFU/g.
        → results in
3. Secretion and accumulation of preformed, HEAT-STABLE enterotoxin(s)
   (SEA predominant) in the food.
        → (cooking/reheating kills bacteria but NOT toxin)
4. Ingestion of toxin-laden food delivers preformed SE to the gut lumen.
        → branches into TWO arms:

   ── ARM A (dominant, emetic) ──────────────────────────────────────
   5a. SE binds intestinal submucosal MAST CELLS
         → induces mast-cell DEGRANULATION
   6a. Release of SEROTONIN (5-HT) and HISTAMINE in small intestine
         → activates 5-HT3 receptors on VAGAL AFFERENT neurons
   7a. Vagal signaling to brainstem emetic center
         → VOMITING, nausea, retching, cramps  (abolished by vagotomy)

   ── ARM B (superantigen, dose-dependent) ──────────────────────────
   5b. SE cross-links MHC class II (APC) × TCR Vβ (T cell)
         → POLYCLONAL T-cell activation (MHC-II/Vβ-restricted, not peptide)
   6b. Cytokine storm: IL-2, IFN-γ, TNF (MAIT cells early, then anergic)
         → systemic inflammation; at high/parenteral dose → TOXIC SHOCK
8. Self-limited resolution in 24–48 h as toxin is cleared (Arm A);
   or severe systemic illness requiring intensive support (Arm B).
```

Upstream vs downstream. The upstream initiating events are entirely *exogenous* (bacterial contamination + temperature abuse + toxin synthesis); there is no host genetic lesion. Within the host, mast-cell degranulation (Arm A) and MHC-II/TCR-Vβ engagement (Arm B) are the proximal upstream events, with 5-HT3/vagal signaling and the cytokine cascade, respectively, as downstream effectors.

Why two arms matter. The marmoset experiment is decisive: classical emetic SEs (SEA/SEB/SEC) cause vomiting, but the superantigen TSST-1 does not — demonstrating that superantigenicity and emesis are *separable* properties of these toxins (PMID: 31112582). In ordinary oral food poisoning, Arm A dominates and the illness is self-limited; Arm B becomes clinically important at high systemic/parenteral exposure (e.g., toxic shock).

Comparative enterotoxin-gene prevalence by reservoir

Source / study	Dominant *se* gene(s)	Key figure	PMID
Human foodborne-outbreak isolates	**SEA**	SEA detection 63.16%	42350958
Spent laying hens (poultry)	**seh** > sed > ser	50.6% carried ≥1 *se* gene; 81.1% birds carried *S. aureus*	42425691
Buffalo subclinical mastitis	**sec** > sea > sed > seb > see	sec 8.5%, sea 8.2%	42406231

Ontology term suggestions

Category	Suggested terms
Disease	MONDO:0005971 (staphylococcal food poisoning)
Phenotype (HPO)	HP:0002013 Vomiting; HP:0002018 Nausea; HP:0002027 Abdominal pain; HP:0002014 Diarrhea
Biological process (GO)	GO:0001821 histamine secretion; GO:0002830 positive regulation of type 2 immune response; GO:0042098 T cell proliferation; GO:0001816 cytokine production; GO:0042310 vasoconstriction (shock arm)
Cell types (CL)	CL:0000097 mast cell; CL:0000624 CD4-positive αβ T cell; CL:0000940 mucosal invariant T (MAIT) cell; CL:0000236 B cell/APC (MHC-II); CL:0002254 enterocyte of small intestine
Anatomy (UBERON)	UBERON:0002108 small intestine; UBERON:0001155 colon; UBERON:0001759 vagus nerve; UBERON:0000945 stomach
Chemical (CHEBI)	CHEBI:28790 serotonin (5-HT); CHEBI:18295 histamine; staphylococcal enterotoxin (protein toxin)

Section-by-Section Characterization

1. Disease information. Acute self-limited emetic food intoxication. Identifiers: MONDO:0005971; MeSH "Staphylococcal Food Poisoning" (D013205); ICD-10 A05.0 (foodborne staphylococcal intoxication). **No OMIM** (not a Mendelian disease). Synonyms: staphylococcal food poisoning, staphylococcal toxemia due to food, staphylococcal enterotoxicosis. Information is derived from **aggregated disease-level** sources (outbreak investigations, surveillance, food-microbiology studies), not individual-patient genetic records (F001, F005).

2. Etiology. Primary cause is **infectious/toxigenic-environmental**: ingestion of preformed *S. aureus* enterotoxins. Risk factors are environmental/behavioral — poor food hygiene, bare-hand contact with ready-to-eat food, temperature abuse, colonized food handlers, mastitic animal milk. **No human genetic risk, protective, or gene-environment factors** apply (F001, F003, F006, F010).

3. Phenotypes. Symptoms/signs: vomiting (dominant), nausea, abdominal cramps, variable non-bloody diarrhea; fever usually absent/low-grade. Onset acute (~1–6 h); severity mild–moderate (self-limited) but occasionally severe with dehydration; course episodic/self-limited over 24–48 h. Frequency: vomiting near-universal; diarrhea variable (F002, F011).

4. Genetic/molecular. "Causal genes" are *bacterial*: *sea*, *seb*, *sec*, *sed*, *see* plus *seg–selu*, *seh*, *ser*, etc., carried on prophages, pathogenicity islands, plasmids, and the enterotoxin gene cluster (*egc*). SEA predominant in human outbreaks. No human HGNC gene, no pathogenic human variant, no ACMG classification, no allele-frequency/gnomAD data applicable. The relevant "variants" are *se*-gene presence/absence and SE serotype (F007).

5. Environmental. Vehicles: handled cooked meats/poultry, dairy/cheese, salads, cream-filled bakery. Reservoirs: human nasal/skin carriage, animal mastitis. Lifestyle factor: food-handling practices (F003, F006).

6. Mechanism. See causal chain above — two arms (emetic mast-cell/5-HT3/vagal reflex; superantigen MHC-II×TCR-Vβ cytokine storm) (F002, F004).

7. Anatomical structures. Primary: small-intestinal mucosa (submucosal mast cells, enterocytes); vagal afferents (UBERON:0001759). In the superantigen/shock arm, systemic lymphoid tissue and vasculature. Bilateral/systemic rather than lateralized (F002).

8. Temporal development. Onset acute, ~0.5–6 h (typically 2–4 h). No chronic stages; self-limited, resolving in 24–48 h; spontaneous remission the rule (F011).

9. Inheritance & population. No inheritance pattern (not genetic). Among the most common foodborne diseases globally; heavily under-reported; point-source outbreaks with high attack rates (e.g., 15%). No sex or ethnic predilection beyond exposure (F005).

10. Diagnostics. Epidemiologic linkage + direct toxin detection (ELISA/immunoassay) in food/supernatant; *se*-gene PCR; WGS + MLST for outbreak typing; emerging aptasensor/biosensor platforms. No host genetic testing. Differential: *Bacillus cereus* emetic toxin (similar short-incubation vomiting), other preformed-toxin intoxications, norovirus (F008).

11. Prognosis. Excellent — self-limited, full recovery in 24–48 h; mortality rare, mainly from dehydration in vulnerable persons or from severe superantigen/toxic-shock presentations (F005, F011, F012).

12. Treatment. Supportive: oral/IV rehydration, electrolyte correction, antiemetics as needed. **Antibiotics not indicated** (toxin-mediated). NCIT suggestions: NCIT:C70962 Fluid Replacement/Rehydration; antiemetic therapy (F012).

13. Prevention. Food hygiene (hand hygiene, glove/utensil use, exclude colonized handlers), temperature control (rapid cooling, cold holding, avoid prolonged warm holding), mastitis management in dairy herds. Experimental food-stage biocontrol (probiotics, bacteriophage). No vaccine (F010).

14. Other species / natural disease. Enterotoxigenic *S. aureus* naturally colonizes/infects poultry (81.1% carriage in spent laying hens), cattle and buffalo (subclinical mastitis with *se* genes), camels, and other food animals — important as reservoirs feeding the human food chain (F006, F009).

15. Model organisms. Non-genetic toxin-challenge emesis models: house musk shrew (*Suncus murinus*), common marmoset (*Callithrix jacchus*), historically rhesus/cynomolgus monkeys and kittens. These recapitulate the emetic phenotype and dissected the mast-cell/5-HT3/vagal pathway; they do not model a human genotype (none exists) (F002, F009).

Evidence Base

PMID	Title (abbrev.)	Role in this report
42700799	*Shenzhen hand-shredded chicken outbreak*	Anchors preformed-toxin etiology, attack rate, preventable control points
41331910	*S. aureus among food handlers (meta-analysis)*	Food-handler reservoir; general etiology statement
17517065	*SE emesis via intestinal 5-HT*	Proves 5-HT3/vagal emetic pathway (vagotomy abolishes emesis)
31112582	*Histamine/mast-cell emesis in marmoset*	Mast-cell degranulation upstream; SE vs TSST-1 separability
41187575	*SEs in cancer immunotherapy*	Defines superantigen MHC-II×TCR-Vβ mechanism and cytokines
30909619	*Pyrogenic toxins induce toxic shock*	Superantigen → toxic shock (Arm B)
28632753	*MAIT cells & superantigen immunopathology*	Early cytokine source then anergy — immunosuppression mechanism
42314251	*Predictive modeling in chicken*	Growth/toxin thresholds; temperature dependence
42190559	*Detection of foodborne protein toxins*	Toxin persists without bacteria → direct detection essential
42418034	*Multi-strain probiotic suppression*	Heat-stable *sea/seb*; experimental biocontrol
36809532	*Staphylococcal toxin in food (protocol/meta)*	Global frequency; under-reporting; short-duration symptoms
42350958	*Genomic/virulence of outbreak isolates*	SEA predominance (63.16%); WGS/MLST diagnostics
42425691	*S. aureus in spent laying hens*	*se*-gene prevalence; poultry reservoir; ELISA method
42406231	*S. aureus buffalo mastitis MLST*	Enterotoxin-gene prevalence in dairy reservoir
42220823	*SFP from cheese products*	Dairy vehicle; "leading foodborne illness"
42625416	*Nonclassical enterotoxin genes, mastitis milk*	Classifies SE types; thermostable superantigen definition

PMID	Title (abbrev.)	Role in this report
42782413	*Phage UHP46 biocontrol in dairy*	Experimental food-stage phage intervention
42967785	*MOF aptasensor for S. aureus*	Emerging rapid detection platform

Evidence source types. Human clinical/epidemiologic: 42700799, 41331910, 36809532, 42220823, 42350958. Animal model (in vivo emesis): 17517065, 31112582. In vitro/mechanistic immunology: 41187575, 30909619, 28632753. Food microbiology / in vitro: 42314251, 42418034, 42190559, 42425691, 42406231, 42782413, 42967785.

Limitations and Knowledge Gaps

- **No patient-level or omics datasets were analyzed.** This disease has no human genetic component; the investigation is necessarily literature-based and the "molecular profiling" sections (patient transcriptomics/proteomics/metabolomics) are not applicable.
- **Clinical incubation/frequency figures (0.5–6 h; vomiting near-universal)** derive from established CDC/clinical-microbiology descriptions rather than from a single quantified cohort in the reviewed set; exact per-symptom frequencies are approximate.
- **Mechanistic emesis data are primarily from animal models** (shrew, marmoset). While concordant and mechanistically compelling, human in-vivo confirmation of the full 5-HT3/vagal chain is inferential.
- **The superantigen/toxic-shock arm** is dose- and route-dependent; its contribution to ordinary oral food poisoning (vs parenteral/high-dose exposure) is not precisely quantified.
- **One cited snippet** (P 30909619) was flagged as a title-level ("mismatch") match rather than an abstract quote; the superantigen-shock claim is independently supported by P 41187575 and 28632753.
- **Quantitative global burden** (incidence per 100,000) is poorly defined because of systematic under-reporting of this short, self-limited illness.

Proposed Follow-up Experiments / Actions

1. **Quantify per-symptom frequencies** from a pooled modern outbreak dataset (vomiting, diarrhea, fever, cramps, incubation distribution) to populate HPO frequency annotations precisely.
2. **Molecular dose–response mapping** of emetic vs superantigen activity across SE serotypes (SEA–SEE, SEH, newer *sel* types) in a standardized marmoset/shrew assay to rank clinical risk by toxin type.
3. **Validate direct-toxin rapid diagnostics** (aptasensor, LC-MS/MS) against gold-standard ELISA in real food matrices for point-of-care outbreak confirmation (PMID: 42967785).
4. **Field-test food-stage biocontrol** (probiotic consortia, phage UHP46) in dairy/poultry supply chains for toxin-formation suppression, with *sea*/*seb* expression readouts (PMID: 42418034; PMID: 42782413).
5. **Strengthen handler-screening and temperature-control interventions** and measure outbreak-rate reduction, targeting the two documented control points (bare-hand contact; inadequate cooling) (PMID: 42700799).
6. **Mechanistic therapeutics**: test whether 5-HT3 antagonists (e.g., ondansetron) shorten symptom duration in human SFP, given the animal-model pathway evidence (PMID: 17517065).

Report compiled from 12 confirmed findings and 34 reviewed papers over 5 investigation iterations. Staphyloenterotoxemia (MONDO:0005971) is a toxin-mediated food intoxication with no human causal gene; its "causal genes" are the bacterial staphylococcal enterotoxin (se) gene family.
